

Examining Dynamic Interdependencies among Global Financial Markets

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Abstract

This paper investigates dynamic interdependencies among major global financial markets from January 1999 to April 2017 by examining their conditional correlations, and risk and return spillovers. Interactions of risk and return are also analysed within the sample markets. Results reveal strong linkages among the global financial markets that intensify during the turmoil period. Within-market spillovers are low indicating their limited exposure to idiosyncratic effects. Further, we find return spillovers of high magnitude onto risk and almost negligible risk spillover onto returns. Spillovers across risk and return fall during the crisis period. These findings are relevant for portfolio management, risk assessment and asset pricing, and have implications for financial markets rating system.

Keywords: Financial markets, conditional correlations, spillovers, conditional volatility, conditional skewness

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1. Introduction

With the advance of globalization, financial markets across the globe have experienced increased linkages over the past few decades. Several factors like arrival of information and communication technologies, development of electronic trading systems and databases, access to information in continuous time and sophisticated marketing tools have contributed to this greater interaction among the markets (Jawadi et al., 2015). Financial markets are thus no more isolated and information originating in one market quickly propagates to another via transmission channels. Assessing financial linkages among the global markets is important as it has implications for international investors as they strive to manage their portfolio and exposure to risk through diversification. Risk and return relationship, therefore, plays a major role on which investors base their decisions for optimal portfolio selection.

Standard asset pricing models postulate that return is linearly related to market-wide volatility risk. In empirical literature, the relationship between return and volatility as a proxy for risk has been extensively studied (for recent literature, see Cotter et al., 2015; Badshah et al., 2016; Chang, 2016; Frazier and Liu, 2016; Kundu and Sarkar, 2016; Jin, 2017; Liu, 2017). However, cross-section of equity returns having substantial exposure to higher moments of returns that are not captured by the volatility risk has been documented in the literature. Chang et al. (2013) show that market skewness risk is negatively related to equity returns and cannot be explained by other common risk factors like size, book-to-market, momentum and market volatility factors. Amaya et al. (2015) compute realized moments for equity returns and confirm significant relation between equity returns and realized skewness. Hence, investigation of interaction between return and risk captured by higher-order moments is warranted.

Extant research has been conducted on examining linkages among international equity markets by analysing risk and return spillovers. These studies typically focus on investigating return or/and volatility spillovers across markets either by applying Multivariate Generalized Autoregressive Conditional Heteroscedasticity (M-GARCH) models or spillover index methodology of Diebold and Yilmaz (2009, 2012, 2014) (see, among others, Bhar and Nikolova, 2009; Singh et al., 2010; Sugimoto et al., 2014; Zhang and Wang, 2014; Celements

et al. 2015; Liow, 2015; Rejeb and Arfaoui, 2016; Kundu and Sarkar, 2016; Fowowe and Shuaibu, 2016; Allen et al., 2017; Liu et al. 2017; Rahahleh and Bhatti, 2017; Shahzad et al., 2017). However, interlinkages between markets can change dramatically during crisis period that extend beyond the risk associated with change in volatility, and include additional crisis transmission channels operating through higher-order moments of asset returns (Fry et al., 2010; Fry-McKibbin et al., 2014). Studies that examine the dynamics of higher-order moment spillovers across markets are scant (see for example, Do et al. 2016, Greenwood-Nimmo et al. 2016). Hence, it is relevant to study interdependencies among markets by incorporating the risk associated with higher-order moments of returns.

This paper aims to investigate dynamic interdependencies among global financial markets by examining their conditional correlations and risk and return spillovers. We base our analysis on volatility and skewness as risk factors¹. Since equity market returns are generally characterized by asymmetric volatility dynamics as has been evidenced by overwhelming literature, we employ Exponential GARCH (EGARCH) Model introduced by Nelson (1991) to compute conditional volatility. We specify conditional skewness as both autoregressive and dependent on lagged equity return shocks to allow for time dependencies in shape of return distribution and to capture the impact of a market shock on the degree of asymmetry, respectively. As first step to the analysis of financial market linkages, we investigate correlation dynamics of the sample equity markets using Multivariate GARCH model framework. As suggested by Cappiello et al. (2006), stock market correlation dynamics are characterized by asymmetries. Also, there has been growing literature that acknowledges presence of asymmetric financial market correlations that become more volatile at the times of joint negative shock based on their empirical evidence (see Kim and Sun, 2017). Hence, we employ Asymmetric Dynamic Conditional Correlation model (ADCC) given by Cappiello et al (2006) to examine conditional correlation dynamics of the sample equity markets. We then turn to employ block aggregation technique under Diebold-Yilmaz framework proposed by Greenwood-Nimmo et al. (2015) to examine risk and return spillovers across the global financial markets. Block aggregation is a novel approach that augments the flexibility of Diebold-Yilmaz framework by integrating a set of individual variables so as to examine spillovers among a group of variables. The methodology, thus, enables us to study spillovers

¹ Following Greenwood-Nimmo et al. (2016), we do not include kurtosis as a measure of risk as it is found to have little empirical relevance. Literature on equity returns and kurtosis risk also has not established a significant relation between the two (see Tang and Shum, 2003; Chang et al., 2013; Amaya et al. 2015).

in a unified framework and provides a richer analysis of risk and return spillover among the global financial markets. Adopting similar aggregation routine under Diebold-Yilmaz framework, we also examine interactions between risk and return within the global financial markets.

Our analysis is carried out on 13 major global financial markets selected from the ranking based on Global Financial Centres Index (GFCI 21²). GFCI is a widely known and accepted index measuring the competitiveness of financial centres worldwide based on five key parameters viz. business environment, financial sector development, infrastructure, human capital and reputation³. The selection of major global financial markets for analysing the risk and return spillovers within and across markets entails two benefits. First, these markets being the hub of investment activities encapsulate bulk of the trading transactions of investor's community. Analysis on these markets thus provides a concise account of risk and return spillovers in the global financial system. Additionally, we are able to evaluate the existing rating system for the financial centres in light of the information transmission across the markets. One would naturally assume that greater competitiveness of financial market based on GFCI would consequently imply its stronger linkages with the global markets thereby creating a global financial network. Our investigation of financial interdependencies among the markets shall therefore elucidate the dominance or strength of a financial market and provide useful insights regarding the global financial market rating system.

Our study makes several important contributions to the literature. We examine dynamic linkages among major global equity markets by studying their conditional correlations and risk and return spillovers. Our study is a significant departure from the literature on equity market linkages as our analysis of information transmission encompasses return and the accompanying risk that investors face, which we capture by conditional volatility and conditional skewness. We also ascertain spillovers among return and higher-order conditional moments to examine interaction between risk and return within the global financial markets. Based upon the results, we draw implications for international investors who are interested in portfolio management and risk assessment, as well as policymakers for policy formulation.

² Global Financial Centres Index (GFCI) 21 report is available from the following web link: http://www.montrealinternational.com/wp-content/uploads/2017/03/gfci_21.pdf

³ GFCI is compiled and published twice a year by Z/Yen Group. The assessment of the competitiveness of a financial centre is based on instrumental factors obtained from external sources and survey of financial services professionals through online questionnaire.

Further, we infer underlying dominance or strength of a financial market and undertake comparative analysis of GFCI ranking and the ranking based on information transmission.

We find that financial linkages among the sample equity markets have limited exposure to idiosyncratic effects as they are largely driven by common global economic factors. The systematic effects in these markets intensify during the turmoil period, pointing to financial contagion. UK is the most dominant market based on information transmission, followed by Australia and South Korea. India and Japan, on the other hand, are found to be least dominant among all sample markets. With regards to interaction between risk and return in global financial markets, results reveal return spillovers of high magnitude onto risk and almost negligible risk spillover onto returns. This indicates that return has greater role to play in volatility prediction than vice-a-versa. Further, we find evidence of risk-return spillovers dampening during the turmoil period which is characterized by large jumps in the asset prices. Systematic effect amplifies during the crisis, highlighting the role of common global factors in propagating risk and return across the global financial markets in the stress times. These findings are relevant for portfolio management, risk assessment and policy formulation. Our results also have implications for Global Financial Centres Index (GFCI) ranking system and asset pricing theories.

The remainder of the paper is structured as follows: Section 2 describes data along with the estimation of the higher-order conditional moments. Section 3 outlines the methodological framework adopted in the paper. Section 4 and 5 present empirical results and discussions. In Section 6, we provide summary and conclusion.

2. Data and Conditional Moments

2.1. Data

The dataset for the study comprises of weekly MSCI equity index closing prices of 13 major global financial markets viz. Australia, Canada, China, Germany, Hong Kong, India, Japan, Luxembourg, Singapore, South Korea, Switzerland, United Kingdom (UK) and United States (US) over the period January 13, 1999 to April 26, 2017, retrieved from Thomson Reuters Eikon. The ranking of global financial centres based on GFCI 21 is taken as reference to select the sample markets for our study. We first consider top 25 financial centres and specify

the financial market in which these centres are based. We then arrive at the condensed list of 13 major global financial markets as the overlapping markets are dropped from the list⁴.

These 13 major financial markets are United Kingdom (UK), United States (US), Singapore, Hong Kong, Japan, Australia, Canada, Switzerland, China, Luxembourg, Germany, South Korea and United Arab Emirates (UAE). UAE, however, is dropped from the analysis because of the unavailability of data since the start of our sample period. We instead consider India among the list of major global financial market for analysis as it is one of the fastest growing emerging markets and has significantly improved its GFCI rating over the past years. Since our analysis is based on international equity markets, we use weekly frequency of equity indices (sampled on Wednesday), instead of daily, to counter the problem of non-synchronous trading bias and day-of-the-week effects. Weekly returns are then computed as the first difference of log-transformed price index series.

2.2. Preliminary analysis of sample equity returns series

Summary statistics of the sample index return series are presented in Table 1. Average return is positive and low, while standard deviation is high for all sample countries. Mean return is highest for India and lowest for UK. All the sample markets are characterized by negative skewness and excess kurtosis. Jarque-Bera test confirms departure of return distributions from normality as the test statistic is found to be significant for all cases. Further, all sample return series are serially correlated (except Australia) and exhibit conditional heteroscedasticity as indicated by Ljung-Box Q-statistic and ARCH Lagrange Multiplier test, respectively. We also perform Augmented Dickey–Fuller (Dickey and Fuller, 1979), KPSS (Kwiatkowski et al., 1992) and Phillips–Perron (Phillips and Perron, 1988) tests as a precursor to time-series analysis, and results reveal that all sample return series are integrated at level.

⁴ For example, New York, San Francisco and Chicago are the three financial centres located in United States and rank 2nd, 6th and 7th according to GFCI 21 rating. Hence, we consider US as a financial market and assign rank 2 to it. Similarly, Japan has two financial centres viz. Tokyo and Osaka ranked 5th and 15th. We thus consider Japan as a financial market for our analysis and place it on rank 5.

Table 1: Summary Statistics of Sample Equity Return Series

Market	Mean	Std. Dev.	Max	Min	Skew	Kurt	JB	LB	ARCH LM	Obsv
AUS	0.0009	0.0321	0.1635	-0.2524	-0.80	9.33	1695.06** [0.00]	18.09 [0.11]	94.70** [0.00]	955
CAN	0.0011	0.0305	0.1013	-0.1944	-0.69	6.33	516.58** [0.00]	33.86** [0.00]	165.24** [0.00]	955
CHI	0.0008	0.0418	0.2089	-0.1956	-0.36	5.96	369.07** [0.00]	25.13** [0.01]	44.14** [0.00]	955
GER	0.0016	0.0451	0.3002	-0.2648	-0.27	8.00	1004.82** [0.00]	24.61* [0.02]	162.56** [0.00]	955
HKG	0.0010	0.0301	0.1403	-0.1580	-0.29	5.77	319.06** [0.00]	28.89** [0.00]	22.90** [0.00]	955
IND	0.0019	0.0403	0.2023	-0.1793	-0.36	5.71	312.86** [0.00]	33.74** [0.00]	31.83** [0.00]	955
JAP	0.0003	0.0280	0.1361	-0.1506	-0.21	4.78	132.95** [0.00]	21.23* [0.04]	30.55** [0.00]	955
LUX	0.0006	0.0298	0.1286	-0.2254	-0.88	8.47	1311.49** [0.00]	35.19** [0.00]	52.79** [0.00]	955
SGP	0.0006	0.0237	0.1034	-0.1675	-0.64	7.91	1023.80** [0.00]	30.42** [0.00]	47.30** [0.00]	955
SKOR	0.0003	0.0356	0.1398	-0.1750	-0.70	5.93	419.37** [0.00]	26.78** [0.00]	45.78** [0.00]	955
SWIZ	0.0006	0.0256	0.1031	-0.1365	-0.47	5.19	225.95** [0.00]	28.11** [0.00]	83.23** [0.00]	955
UK	0.0002	0.0244	0.1359	-0.1273	-0.33	6.40	475.79** [0.00]	24.30* [0.02]	105.21** [0.00]	955
US	0.0009	0.0304	0.1839	-0.1705	-0.11	6.69	544.44** [0.00]	20.72* [0.05]	17.29** [0.00]	955

Notes: (a) Std. Dev., Max, Min, Skew and Kurt denote Standard Deviation, Maximum, Minimum, Skewness and Kurtosis of the return series.

(b) JB denotes Jarque-Bera test for the null hypothesis of normal distribution.

(c) LB denotes Ljung-Box Q-statistic reported at 12th lag and ARCH LM denotes ARCH Lagrange Multiplier test of conditional heteroscedasticity.

(d) Obsv denotes number of observations in the return series.

(e) Value in [] includes p-value

(f) **/* indicates significance at 0.01/0.05 level

2.3. Computation of Conditional Moments

For our analysis, conditional volatility and conditional skewness are used as a measure to capture risk that investors face while making investment decisions. We first begin by specifying conditional mean equation for equity returns of sample markets as an autoregressive process of order 1:

$$r_{i,t} = \mu + \phi r_{i,t-1} + \epsilon_{i,t}, \text{ where } \epsilon_{i,t} \text{ is the disturbance term.}$$

We then compute time-varying conditional volatility by specifying univariate GARCH process as our all sample return series exhibit conditional heteroscedasticity. Exponential GARCH model of Nelson (1991) is used for specification of conditional volatility to avoid using non-negativity restrictions on GARCH parameters and to capture asymmetric response to return innovations. EGARCH(1,1)⁵ model is given as:

$$\ln(h_{i,t}^2) = \omega + \beta \ln(h_{i,t-1}^2) + \alpha \left| \frac{\epsilon_{i,t-1}}{h_{i,t-1}} \right| + \gamma \frac{\epsilon_{i,t-1}}{h_{i,t-1}}$$

Following Sehgal et al. (2015), we obtain time-varying conditional skewness by first estimating skewness parameters using rolling window of 12 weeks (approximately 3 months) for all sample return series. Time-varying skewness is then specified as both autoregressive (AR(1)) and dependent on lagged return shocks (standardized residual) to allow for time dependencies in shape of return distribution and to capture the impact of a market shock on the degree of asymmetry, respectively. Conditional skewness is obtained from the fitted values of following specification:

$$\lambda_{i,t} = \beta_{i,0} + \beta_{i,1} \lambda_{i,t-1} + \beta_{i,2} \epsilon_{i,t-1}, \text{ where } \lambda_{i,t} \text{ is the time-varying skewness parameter.}$$

3. Methodology

3.1. Asymmetric Dynamic Conditional Correlation Model

We apply Asymmetric Dynamic Conditional Correlation (ADCC) model proposed by Cappiello, Engle and Sheppard (2006) to examine time-varying conditional correlations among global financial markets. ADCC model estimates time-varying correlations by taking

⁵ EGARCH(1,1) is chosen as the preferred model in interest of parsimony of parameters (see Kim and Wang, 2006).

into account conditional asymmetries in correlation dynamics that are observed to increase more following joint negative shock than joint positive shock.

Asymmetric dynamic conditional correlations in a multivariate framework are estimated following a two-step procedure. Firstly, univariate volatility for each return series is modelled using Exponential GARCH (EGARCH) process as it accounts for asymmetries in conditional variances of returns and also doesn't impose non-negativity restrictions on GARCH parameters (see Nelson, 1991). The mean equation is specified as an AR(1) process and variance equation as an EGARCH (1,1) process in terms of conditional variance of returns ($h_{i,t}$).

Conditional covariance matrix H_t is decomposed as $H_t = D_t R_t D_t$, where $D_t = \text{diag}\{\sqrt{h_{i,t}}\}$ is $n \times n$ diagonal matrix of time-varying standard deviations for the returns that are extracted from EGARCH (1,1) process and $R_t = \{\rho_{ij}\}_t$ is $n \times n$ time-varying correlation matrix.

After obtaining the estimates of $h_{i,t}$ from EGARCH(1,1) model, standardized residuals i.e. returns transformed by their estimated standard deviation ($\varepsilon_{i,t} = r_{i,t}/\sqrt{h_{i,t}}$) are used to estimate dynamic conditional correlation parameters in the second step. The asymmetric dynamic correlation structure of Cappiello et al. (2006) is given as:

$$Q_t = (\bar{Q} - a^2 \bar{Q} - b^2 \bar{Q} - g^2 \bar{N}) + a^2 (\varepsilon_{t-1} \varepsilon'_{t-1}) + b^2 Q_{t-1} + g^2 (\eta_{t-1} \eta'_{t-1}),$$

where Q_t is positive definite if the matrix in parentheses is positive semi-definite, $\bar{Q} = E(\varepsilon_t \varepsilon'_t)$ is $n \times n$ unconditional correlation matrix of standardized residuals, $\bar{N} = E(\eta_t \eta'_t)$ and g is the asymmetric term which captures the period of joint negative shock. For Q_t to be positive definite, the matrix in parentheses should positive semi-definite.

The dynamic correlation matrix is written as:

$$P_t = Q_t^{*-1} Q_t Q_t^{*-1},$$

where $Q_t^* = [\sqrt{q_{iit}}]$ is a diagonal matrix with the square root of the i th diagonal elements of Q_t on its diagonal entries.

We choose ADCC model over other multivariate GARCH models because of its several advantages. ADCC model parametrizes conditional correlations, while other specifications like VEC and BEKK require indirect computations of conditional correlations as they parametrize conditional co-variances (Gjika and Horvath, 2013). Constant Conditional

Correlation (CCC) model introduced by Bollerslev (1990) assumes correlations to be constant which is neither realistic nor empirically justified (Gjika and Horvath, 2013) and its estimation involves solving a high dimensional optimization problem (Aboura and Chevallier, 2014). Dynamic Conditional Correlation (DCC) model proposed by Engle (2002) relaxes the assumption of constant correlation and provides flexibility by reducing the number of parameters to estimate (see Kim and Sun, 2017). However, ADCC model is preferred as it nests DCC model (Syriopoulos and Roumpis, 2009) and incorporate asymmetries in conditional correlations.

3.2. Spillover index based on block aggregation

Spillover index methodology proposed by Diebold and Yilmaz (2009, 2012) which is based on vector autoregressive (VAR) framework allows us to examine spillovers across variables. It quantifies the contribution of shocks to and from each variable in terms of each variable's forecast error variance through variance decomposition analysis, and therefore provides the magnitude and direction of spillovers. Diebold and Yilmaz (2012) uses generalized VAR framework of Koop et al. (1996) and Pesaran and Shin (1998) which yields forecast-error variance decompositions that are invariant to the ordering of the variables.

The N variable VAR of p^{th} order can be written as: $y_t = \sum_{i=1}^p \phi_i y_{t-i} + \varepsilon_t$, where $y_t = (y_{1t}, \dots, y_{Nt})$ is a vector of N endogenous variables, ϕ_i is $N \times N$ parameter matrix and $\varepsilon_t \sim (0, \Sigma)$ is a vector of innovations. Its moving average representation can be written as $y_t = \sum_{i=1}^{\infty} A_i \varepsilon_{t-i}$, where $A_i = \sum_{j=1}^p \phi_j A_{i-j}$ and $j=1, \dots, p$.

The H -step ahead forecast error variance decomposition of i^{th} variable which can be attributed to shocks for j^{th} variable is given as:

$$\theta_{i \leftarrow j}^g(H) = \frac{\sigma_{ii}^{-1} \sum_{h=0}^{H-1} (e_i' A_h \Sigma e_j)^2}{\sum_{h=0}^{H-1} (e_i' A_h \Sigma A_h' e_j)}, \text{ for } i, j = 1, \dots, N$$

where Σ is the estimated variance matrix for the error term of VAR, σ_{ij} is the standard deviation of the error term for the i^{th} equation and e_i is the selection vector with one for the i^{th} element and zero otherwise. Each forecast error variance decomposition is normalized by the

$$\text{row sum as: } \theta_{i \leftarrow j}^g(H) = \frac{\theta_{i \leftarrow j}^g(H)}{\sum_{j=1}^N \theta_{i \leftarrow j}^g(H)}$$

$N \times N$ connectedness matrix can thus be constructed using the forecast error variance decompositions as follows:

$$C(H) = \begin{bmatrix} \theta_{1 \leftarrow 1}^g(H) & \theta_{1 \leftarrow 2}^g(H) & \cdots & \theta_{1 \leftarrow N}^g(H) \\ \theta_{2 \leftarrow 1}^g(H) & \theta_{2 \leftarrow 2}^g(H) & \cdots & \theta_{2 \leftarrow N}^g(H) \\ \vdots & \vdots & \ddots & \vdots \\ \theta_{N \leftarrow 1}^g(H) & \theta_{N \leftarrow 2}^g(H) & \cdots & \theta_{N \leftarrow N}^g(H) \end{bmatrix},$$

where $\theta_{i \leftarrow j}^g(H)$ measures the pairwise spillover from variable j to variable i .

While Diebold-Yilmaz framework provides the measure of pairwise directional spillovers among individual markets, it does not quantify the spillovers among a group of variables. Greenwood-Nimmo, Nguyen and Shin (2015) extend Diebold-Yilmaz framework by exploiting block aggregation of the connectedness matrix which applies an aggregation routine to group a set of individual variables.

We, therefore, adopt the same methodology to examine linkages amongst the global financial markets. We examine the linkages among $N=13$ markets wherein each market encompasses three variables- return, conditional volatility and conditional skewness, in a similar manner as in Greenwood-Nimmo, Nguyen and Shin (2016). The variables are arranged in the order $y_t = (r_{1t}, v_{1t}, s_{1t}, r_{2t}, v_{2t}, s_{2t}, \dots, r_{Nt}, v_{Nt}, s_{Nt})$.

Generalized VAR framework adopted in Diebold and Yilmaz (2012) ensures that forecast-error decomposition is not sensitive to the variable re-ordering and hence supports any desired block structure. Therefore, the connectedness matrix is arranged in the following form:

$$C(H) = \begin{bmatrix} B_{1 \leftarrow 1}^g(H) & B_{1 \leftarrow 2}^g(H) & \cdots & B_{1 \leftarrow N}^g(H) \\ B_{2 \leftarrow 1}^g(H) & B_{2 \leftarrow 2}^g(H) & \cdots & B_{2 \leftarrow N}^g(H) \\ \vdots & \vdots & \ddots & \vdots \\ B_{N \leftarrow 1}^g(H) & B_{N \leftarrow 2}^g(H) & \cdots & B_{N \leftarrow N}^g(H) \end{bmatrix},$$

$$\text{where } B_{i \leftarrow j}^g(H) = \begin{bmatrix} \theta_{r_i \leftarrow r_j}^g(H) & \theta_{r_i \leftarrow v_j}^g(H) & \theta_{r_i \leftarrow s_j}^g(H) \\ \theta_{v_i \leftarrow r_j}^g(H) & \theta_{v_i \leftarrow v_j}^g(H) & \theta_{v_i \leftarrow s_j}^g(H) \\ \theta_{s_i \leftarrow r_j}^g(H) & \theta_{s_i \leftarrow v_j}^g(H) & \theta_{s_i \leftarrow s_j}^g(H) \end{bmatrix}, \text{ for } i, j = 1, 2, \dots, N$$

Total within market forecast error variance contribution for market i is given as:

$$W_{i \leftarrow i}^g(H) = \frac{1}{m} \cdot e_m' \cdot B_{i \leftarrow i}^g(H) \cdot e_m,$$

and the total Pairwise directional spillover from market j to market i ($i \neq j$) at horizon H is given as:

$$P_{i \leftarrow j}^g(H) = \frac{1}{m} \cdot e'_m \cdot B_{i \leftarrow j}^g(H) \cdot e_m,$$

where m is the number of variables that each group is composed of (in this case, $m=3$) and e_m is $m \times 1$ vector of ones.

Hence, the aggregated connectedness matrix following Greenwood-Nimmo et al. (2015) can be written as:

$$C(H) = \begin{bmatrix} W_{1 \leftarrow 1}^g(H) & P_{1 \leftarrow 2}^g(H) & \cdots & P_{1 \leftarrow N}^g(H) \\ P_{2 \leftarrow 1}^g(H) & W_{2 \leftarrow 2}^g(H) & \cdots & P_{2 \leftarrow N}^g(H) \\ \vdots & \vdots & \ddots & \vdots \\ P_{N \leftarrow 1}^g(H) & P_{N \leftarrow 2}^g(H) & \cdots & W_{N \leftarrow N}^g(H) \end{bmatrix}$$

Now, total within-market contribution, $W_{i \leftarrow i}^g(H)$, can be decomposed into common-variable forecast error variance contribution within-market i , $O_{i \leftarrow i}^g(H)$, and cross-variable effects, $C_{i \leftarrow i}^g(H)$, which are given as follows:

$$O_{i \leftarrow i}^g(H) = \frac{1}{m} \text{trace}(B_{i \leftarrow i}^g(H)) \quad \text{and} \quad C_{i \leftarrow i}^g(H) = W_{i \leftarrow i}^g(H) - O_{i \leftarrow i}^g(H)$$

It should be emphasized here that $O_{i \leftarrow i}^g(H)$ is the proportion of forecast error variance of $y_{i,t}$ that is not attributable to spillovers among moments within market i nor to the spillovers from other markets to market i . On the other hand, $C_{i \leftarrow i}^g(H)$ is the proportion of forecast error variance of $y_{i,t}$ that is ascribed to spillovers among moments within-market i .

We can also decompose total pairwise directional spillovers into common-variable, $O_{i \leftarrow j}^g(H)$, and cross-variable effects, $C_{i \leftarrow j}^g(H)$, expressed as:

$$O_{i \leftarrow j}^g(H) = \frac{1}{m} \text{trace}(B_{i \leftarrow j}^g(H)) \quad \text{and} \quad C_{i \leftarrow j}^g(H) = W_{i \leftarrow j}^g(H) - O_{i \leftarrow j}^g(H), \text{ where } i \neq j$$

Here, $O_{i \leftarrow j}^g(H)$ measures the proportion of common-variable forecast error variance of $y_{i,t}$ that is attributable to spillovers from other markets to market i , while $C_{i \leftarrow j}^g(H)$ captures the proportion of forecast error variance of $y_{i,t}$ that is ascribed to spillovers among moments from other markets to market i .

Total directional spillover of market i to/from all other markets in the model can also be estimated from the connectedness matrix. Total directional spillovers transmitted by market i *from* and *to* all other markets, in other words, the aggregate *from* and *to* connectedness of market i are expressed as:

$$F_{i\leftarrow\bullet}^g(H) = \sum_{j=1}^N P_{i\leftarrow j}^g(H) \quad \text{and} \quad T_{i\leftarrow\bullet}^g(H) = \sum_{j=1}^N P_{j\leftarrow i}^g(H), \text{ respectively.}$$

Gross and Net directional spillover of market i can be obtained as follows:

$$G_i^g(H) = T_{i\leftarrow\bullet}^g(H) + F_{i\leftarrow\bullet}^g(H) \quad \text{and} \quad N_i^g(H) = T_{i\leftarrow\bullet}^g(H) - F_{i\leftarrow\bullet}^g(H), \text{ respectively.}$$

The block aggregation technique under Diebold-Yilmaz framework described above is in context of analyzing linkages among markets by aggregating moments. In the same vein, linkages among moments can be examined by aggregating markets. The variables are reordered in the form $y_t = (r_{1t}, r_{2t}, \dots, r_{Nt}, v_{1t}, v_{2t}, \dots, v_{Nt}, s_{1t}, s_{2t}, \dots, s_{Nt})$ such that returns for all markets, conditional volatility of all markets and conditional skewness of all markets are placed in succession. Block structure can be expressed with $N=3$ groups, composed of $m=13$ variables or markets. We then apply aggregation routine in similar manner to examine spillovers among moments.

4. Interdependencies among global financial markets

4.1. ADCC Model Results

We employ ADCC model to investigate linkages among the markets over the sample period. The results of EGARCH (1,1) modelling, employed as the first step to estimate inter-temporal conditional correlation in ADCC framework, are displayed in Panel A of Table 2. Estimation results show presence of long run volatility persistence as is indicated by significant value of β coefficient. Both ARCH (α) and GARCH (γ) term measuring size and leverage effect, respectively are also found to be significant for all sample markets. Negative and significant value of γ coefficient indicates asymmetric effect of news on volatility factor which increases more after a negative shock than a positive shock, thereby justifying the use of EGARCH model. Further, the coefficient estimates a^2 , b^2 and g^2 that pertain to the ADCC equation are found to be significant (refer Panel B of Table 2) indicating that the data fits the multivariate model well. Positive and significant value of g^2 implies asymmetric movements of conditional correlations which are triggered by a joint negative shock.

Table 2: EGARCH and ADCC Model Estimation Results

Panel A: EGARCH(1,1) Model						
Market	Mean Eqn		Variance Eqn			
	μ	$r_i (-1)$	ω	α	γ	β
AUS	0.0009 (0.00)	-0.0213 (0.04)	-0.3102* (0.14)	-0.0978** (0.03)	0.2028** (0.04)	0.9562** (0.02)
CAN	0.0014 (0.00)	-0.0312 (0.03)	-0.3332** (0.10)	-0.0784** (0.02)	0.2291** (0.04)	0.9536** (0.01)
CHI	0.0014 (0.00)	0.0305 (0.03)	-0.2003** (0.08)	-0.0506* (0.03)	0.2556** (0.04)	0.9697** (0.01)
GER	0.0015 (0.00)	-0.0056 (0.04)	-0.2244** (0.07)	-0.0894** (0.03)	0.2342** (0.04)	0.9655** (0.01)
HKG	0.0011 (0.00)	0.0511 (0.03)	-0.2430** (0.06)	-0.0793** (0.03)	0.1994** (0.03)	0.9660** (0.01)
IND	0.0014 (0.00)	0.0763* (0.03)	-0.02372* (0.12)	-0.0492 (0.03)	0.2416** (0.05)	0.9637** (0.02)
JAP	0.0001 (0.00)	-0.0271 (0.04)	-0.887* (0.40)	-0.1259** (0.05)	0.2037** (0.05)	0.8769** (0.06)
LUX	0.0013** (0.00)	0.0283** (0.00)	-0.6139** (0.25)	-0.1354** (0.04)	0.2988** (0.06)	0.9144** (0.03)
SGP	0.0006 (0.00)	-0.2111** (0.03)	-0.4825** (0.01)	-0.4825** (0.03)	0.1140** (0.02)	0.9380** (0.00)
SKOR	0.0010 (0.00)	-0.0653 (0.04)	-0.5144** (0.07)	-0.1562** (0.03)	0.1841** (0.03)	0.9254** (0.01)
SWIZ	0.0005* (0.00)	-0.0775** (0.01)	-0.5962 (0.40)	-0.1560** (0.05)	0.1901** (0.05)	0.9205** (0.05)
UK	-0.0002 (0.00)	-0.0459 (0.04)	-0.6078** (0.05)	-0.2256** (0.03)	0.1458** (0.03)	0.9211** (0.01)
US	0.0007 (0.00)	0.1114** (0.03)	-0.1835** (0.06)	-0.0734** (0.02)	0.2013** (0.03)	0.9743** (0.01)
Panel B: ADCC Model						
ADCC Eqn	a^2		b^2		g^2	
	0.0103** (0.00)		0.9666** (0.01)		0.0067** (0.00)	

Notes: (a) Panel A and Panel B of the table presents parameter estimates of EGARCH (1,1) and ADCC model, respectively.

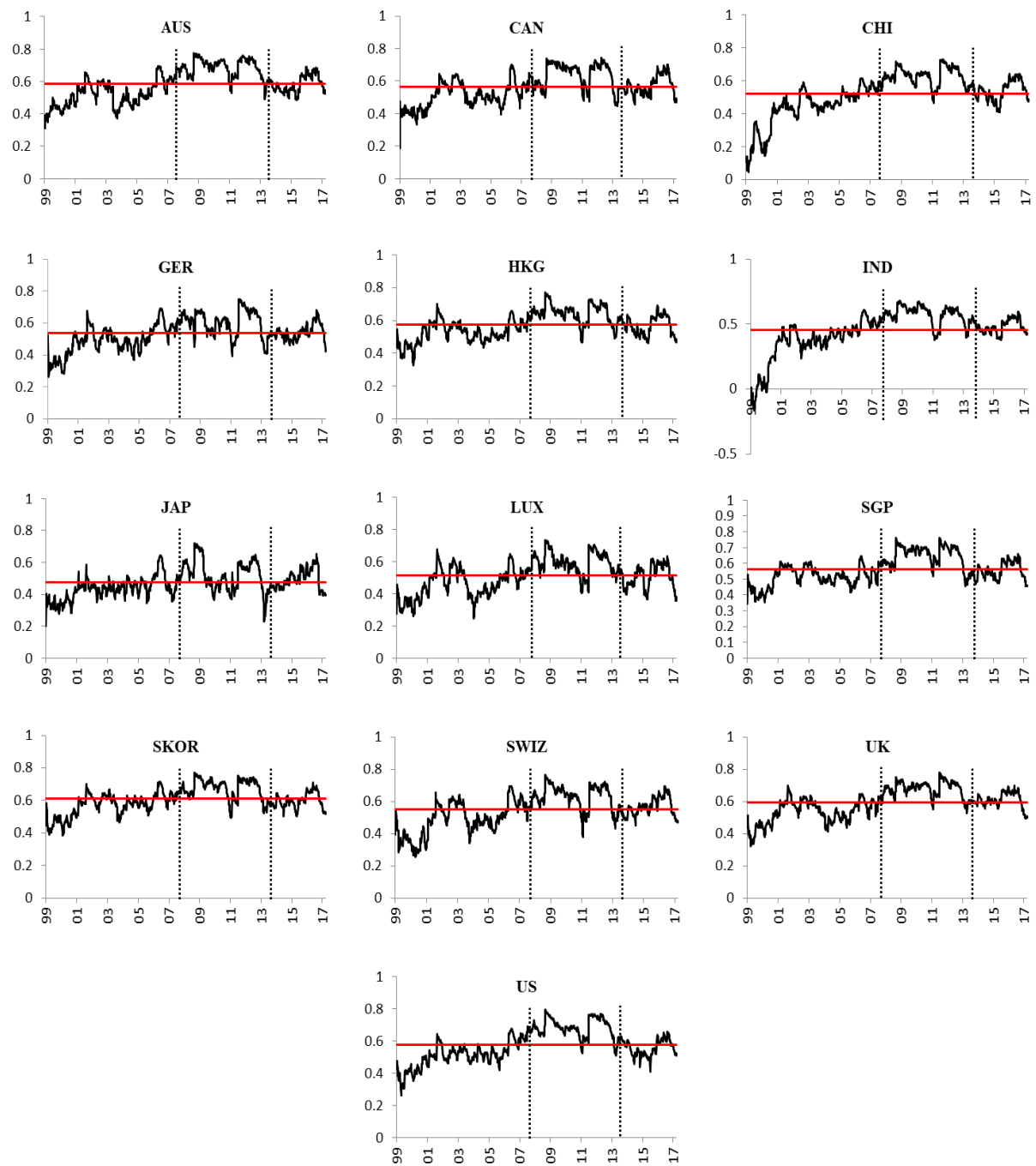
(b) Value in () includes standard error

(c) **/* indicates significance at 0.01/0.05 level

Figure 1 plots the dynamic conditional correlation of each global financial market with rest of the sample markets on an average⁶. The figure also charts out mean value of the conditional correlation of each sample equity market with other markets corresponding to its dynamic correlations (shown as red line). South Korea has the highest average conditional correlation with other sample market (0.61), followed by UK (0.60) and Australia (0.59), implying its strong linkages with other global financial markets. On the other hand, equity markets of India and Japan are found to have weakest linkages with the other countries (0.45 and 0.47, respectively). The dynamic evolution of conditional correlation is fluctuating over the entire sample period and the pattern of time variations are roughly consistent across all the markets. In addition, an increasing trend is observed in conditional correlations for majority of the markets with other global financial centres. This is especially so in case of India and China as they have gradually liberalized their economies and financial sector by introduction of market oriented reforms over the past two decades. Another notable observation regarding the time series behaviour of conditional correlations from the figure reveals that they undergo steep rise in the aftermath of Global Financial Crisis of 2007-08 and The Eurozone Debt Crisis of 2009. This is in conjunction with the previous literature that highlights marked increase in the interdependencies among equity markets during turbulent times, pointing to global financial contagion.

⁶ Pairwise dynamic conditional correlations are not presented due to brevity of space, but are available on request.

Figure 1: Dynamic Conditional Correlations of Global Financial Markets



Notes: (a) The figure depicts dynamic conditional correlation of each global financial market with rest of the sample markets on an average, computed from ADCC model.

(b) Horizontal line (in red colour) shows the mean value of the conditional correlation of each sample equity market with other markets corresponding to its dynamic correlations.

(c) Time period between the two dotted lines depict the crisis period. Global Financial Crisis starts from August 09, 2007 till October 18, 2009 (see Trichet, 2010, Angelini et al., 2011, Sehgal et al., 2016) and the European Sovereign Debt Crisis begins from October 19, 2009 to June 30, 2013 (see, Ahmad and Sehgal, 2014, Sehgal et al., 2016)

4.2. Block Aggregation Results under Diebold-Yilmaz Framework

Though ADCC model provides us a fair estimate regarding the magnitude of linkages among the markets, it remains silent on the direction of interdependencies among them. In this light, Diebold and Yilmaz (2009, 2014) spillover index methodology can help us gain further insights into the financial linkages among the global markets. Application of this methodology to our dataset which comprises of return, conditional volatility and conditional skewness of 13 sample markets yields a 39×39 connectedness matrix representing the linkages across the three moments of each market⁷. However, it is practically not feasible to interpret 1521 elements of the matrix and undertake inter-temporal analysis. Greenwood-Nimmo et al. (2015) bring forth a novel method in which block aggregation approach is used under Diebold-Yilmaz framework to integrate a set of individual variables so as to examine linkages among a group of variables. Therefore, to capture the spillover effects that channel through return, volatility and skewness across the sample markets, we aggregate the three moments by applying block aggregation routine to the 39×39 matrix⁸ representing pairwise directional spillover indices. We supplement the static spillover results with rolling window analysis⁹ (using a 200-week rolling window) to capture the time-varying characteristics of the spillover indices. In the same vein, we aggregate the sample markets to investigate linkages among moments (discussed in Section 5).

Table 3 presents 13×13 market connectedness matrix depicting the spillovers among the sample markets aggregated over the three moments (return, conditional volatility and conditional skewness). The diagonal elements of the matrix represent within market spillovers and off-diagonal elements represent cross-market spillovers. The last column of the table 3 labelled ‘contribution from others’ sums the directional spillovers to market i from rest of the sample markets and the last row labelled ‘contribution to others’ represents the directional spillover from market i to other markets in the model. Total spillover index (bottom right corner of the table) is approximately 66 percent, indicating that the interdependencies among the financial markets are high. Substantial magnitude of cross-market spillovers is also indicated by the low level of within market spillovers. South Korea

⁷ For full-sample analysis, lag length is selected by minimising Schwarz Information Criterion and forecast horizon is set to $H=4$ weeks.

⁸ The 39×39 connectedness matrix obtained from Diebold and Yilmaz (2014) spillover index methodology is not presented due to brevity of space, but is available on request.

⁹ We use a fixed rolling window of 200 weeks for time-varying analysis. We could not estimate with shorter rolling-window width as time dimension of a shorter subsample (75, 100 or 150 weeks) is insufficient with respect to cross-sectional dimension to undertake rolling estimation.

and UK are the two largest gross receiver of spillovers from other global financial markets (accounting to 70.24% and 70.19% of the forecast error variance, respectively). UK, on the other hand, is the largest information transmitter to rest of the markets (84.19%), followed by South Korea (78.75%). India and Japan are least influential in transmitting information to other markets, and are also least responsive to the information originating in the major global financial markets. This corroborates ADCC results that demonstrate highest conditional correlation of South Korea and UK with other countries and lowest for India and Japan.

Table 3: Spillovers among Global Financial Markets

	AUS	CAN	CHI	GER	HKG	IND	JAP	LUX	SGP	SKOR	SWIZ	UK	US	Contri. <i>from</i>
AUS	34.68	8.17	4.18	4.31	4.83	3.33	3.42	6.11	4.87	7.09	6.05	6.35	6.61	65.32
CAN	9.44	33.18	3.61	3.55	4.16	2.87	3.25	6.75	9.63	6.36	4.93	7.12	5.13	66.80
CHI	4.97	5.06	37.97	5.55	13.78	4.13	2.39	3.98	4.27	3.00	2.72	4.48	7.70	62.03
GER	6.52	5.51	5.92	34.83	5.99	4.45	3.88	5.85	6.28	5.08	3.15	4.89	7.68	65.19
HKG	5.54	4.89	11.64	5.48	30.90	4.28	3.32	6.14	4.88	4.67	3.33	5.62	9.29	69.10
IND	5.22	4.05	4.92	7.47	6.78	41.27	2.95	5.48	4.07	3.77	2.55	4.60	6.90	58.74
JAP	5.99	5.56	3.08	4.31	4.23	3.12	40.98	4.90	6.08	5.96	4.11	5.88	5.81	59.02
LUX	5.95	5.58	3.11	3.68	4.84	3.66	2.88	38.64	5.85	7.77	4.94	8.25	4.86	61.37
SGP	6.17	9.94	3.15	4.48	4.34	3.33	3.13	6.54	30.03	6.98	5.95	10.78	5.18	69.96
SKOR	7.35	6.93	2.66	4.02	3.84	2.94	3.75	6.78	6.76	29.75	10.92	9.44	4.86	70.24
SWIZ	7.16	5.82	2.62	3.46	3.85	2.38	2.90	5.66	7.28	13.52	30.18	10.36	4.80	69.81
UK	6.04	7.02	3.09	3.52	4.73	2.97	2.92	7.37	9.91	8.70	8.75	29.81	5.17	70.19
US	7.44	5.25	6.28	6.55	8.78	4.20	3.46	5.40	5.61	5.86	3.67	6.42	31.09	68.92
Contri. to	77.79	73.78	54.26	56.38	70.16	41.66	38.25	70.95	75.47	78.75	61.07	84.19	73.98	65.90

Notes: (a) The 13×13 matrix in the table presents spillovers among the global financial markets obtained by applying block aggregation technique of Greenwood-Nimmo et al. (2015) under Diebold-Yilmaz framework.

(b) The diagonal elements of the matrix represent within market spillovers and off-diagonal elements represent cross-market spillovers.

(c) The last column of the table, 'contri from', represents directional spillovers *from* other markets to market *i*.

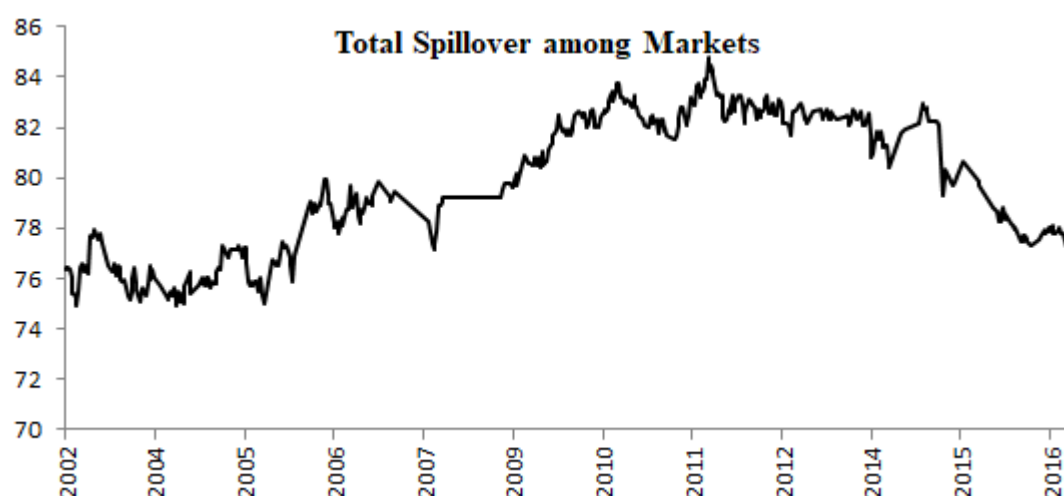
(d) The last row of the table, 'contri to', represents directional spillovers *to* other markets from market *i*.

(e) The bottom right corner of the table is the total spillover index.

(f) All values are measured in percentage units.

Figure 2 depicts the total spillover across the global financial centres based on rolling sample analysis. Substantial time-variation is apparent from the figure as there is a marked increase in the spillover effect since the onset of Global Financial Crisis of 2008. The spillover effect remains pronounced till the end of 2012 and starts falling thereafter, which coincides with the end of Eurozone Debt crisis and beginning of global economic recovery. This is consistent with ADCC results which suggest that the crisis period notably intensified average correlations of majority of the sample markets with rest of the markets.

Figure 2: Total Spillover among Markets

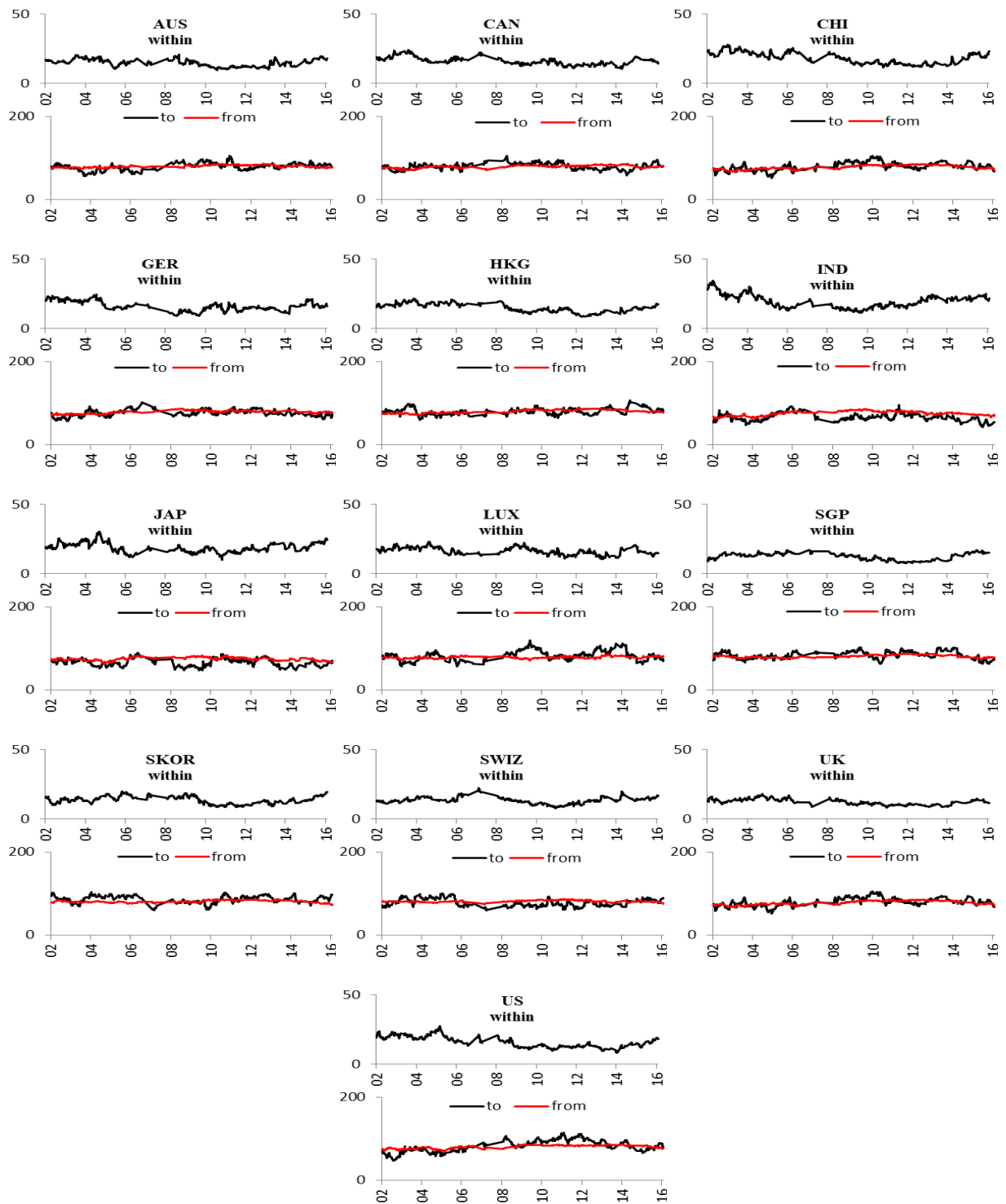


Notes: (a) The figure illustrates evolution of total spillover among global financial markets computed from block aggregation technique under Diebold-Yilmaz framework.

(b) Dynamic spillover index is obtained using fixed rolling window of 200 weeks.

The time evolution of within and directional spillovers of all sample markets is presented in Figure 3. The upper plot of each sample market displays within-market spillovers and the lower plot shows directional spillover ‘to’ and ‘from’ other markets in the sample. Within market spillovers for all sample markets are less than 40% indicating limited exposure to idiosyncratic effects in these markets. This also suggests that there are substantial linkages among the global financial markets as they are driven by common global economic factors implying lack of diversification opportunities. Time variation reveals plunge in idiosyncratic effects for all markets following the outbreak of Global Financial Crisis and gradually increase as the effect of Eurozone Debt crisis subsides in 2012, demonstrating that systematic effects become more pronounced during the turmoil period. India and Japan provide considerable diversification opportunities to international investors as within market effect is relatively strong compared to other global financial markets. Directional spillovers of each

Figure 3: Directional Spillovers among Markets



Notes: (a) The figure illustrates time evolution of within and directional spillovers of all sample markets computed from block aggregation technique under Diebold-Yilmaz framework.
(b) The upper plot of each sample market displays within-market spillovers and the lower plot shows directional spillover 'to' and 'from' other markets in the sample.
(c) Dynamic spillover index is obtained using fixed rolling window of 200 weeks.

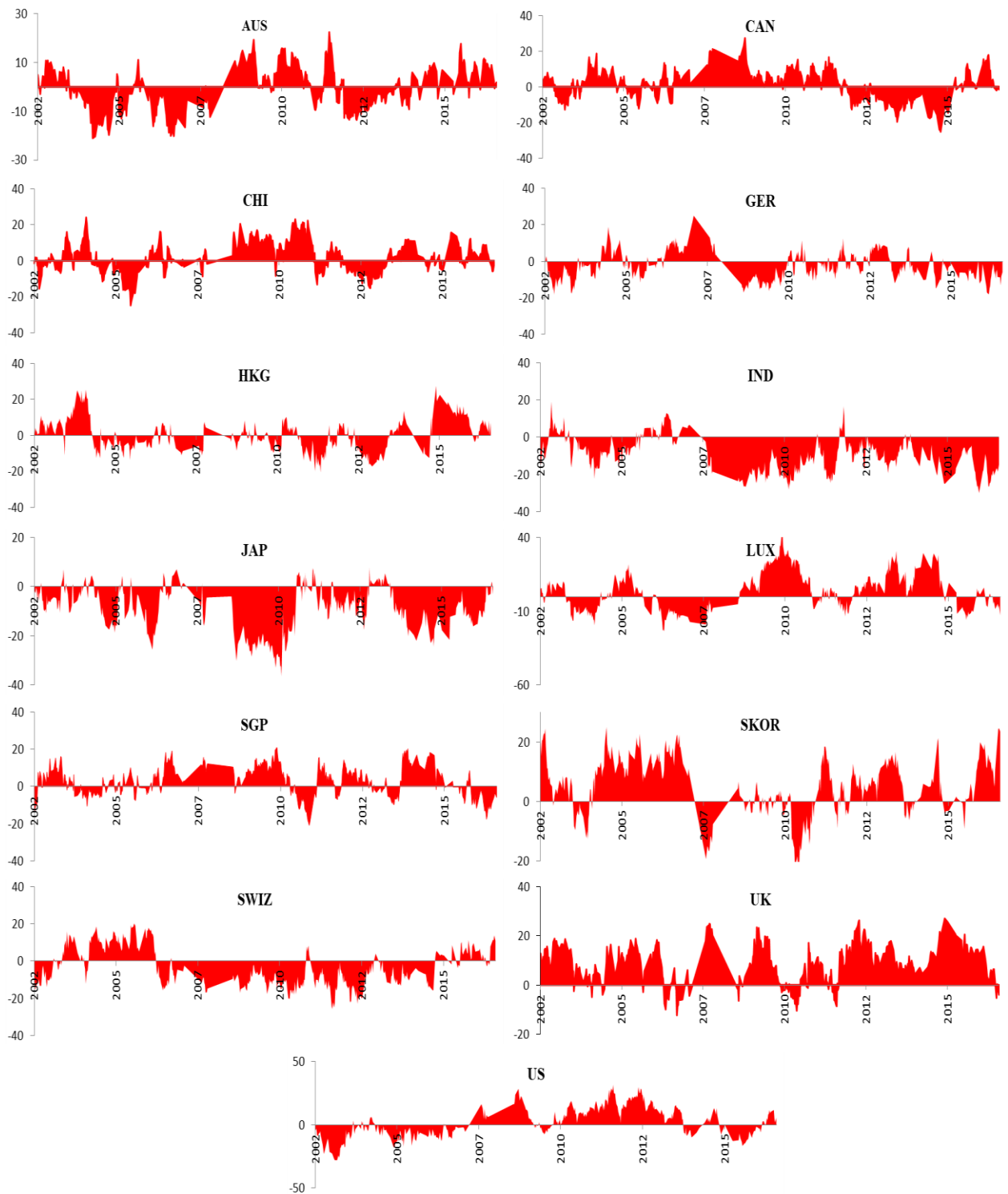
equity market from other markets in the system remain considerably stable over time as is revealed from a little time variation in the evolution of directional spillovers ‘from’ others. On the other hand, directional spillovers ‘to’ others exhibit significant fluctuations in behaviour over the sample period. Spillovers from equity markets of Australia, Canada, China, Luxembourg, UK and US to other markets rise during the crisis period. Rest of the markets experience a decrease, more notably India and Japan. This indicates their weak interdependencies with the global equity markets.

4.3. Inferring dominance of a financial market based on information spillovers

Directional spillovers ‘to’ and ‘from’ other markets are the two key aspects based on the combined effect of which dominance of a financial market can be inferred. Dominance of a financial market will be established if it is influential in transmitting information to other markets, but is relatively less influenced from them. Hence, analysing net spillovers of the financial markets, calculated as the difference between ‘contribution to others’ and ‘contribution from others’ (refer Table 3), is important to gauge their position on a global financial system. Nevertheless, it is pertinent to exercise prudence while drawing inferences based solely on the net spillover index of the markets as it may give spurious picture regarding the dominance of a market if the extent of spillovers ‘to’ and ‘from’ others is low. Therefore, it is of particular interest to decipher the net spillover results in light the gross spillover of the market (calculated as the total of ‘contribution to others’ and ‘contribution from others’). From our results, UK emerges as the lead net transmitter of information to other financial markets along with exhibiting the highest gross spillover activity, and therefore can be regarded as the dominant market in the global financial system based on information transmission. Australia and South Korea are other major sources of spillover indicating that they also exert substantial influence on other markets. Luxembourg, though is a strong net transmitter, cannot be ascribed as one of the dominant markets as the degree of spillovers ‘to’ and ‘from’ others is comparatively low. On the other hand, Japan and India have the lowest gross spillovers besides being the largest net receivers of the spillover from other markets in the system indicating that they are least dominant of all the sample markets.

Figure 4 displays the net directional spillover indices computed through rolling analysis. UK is consistently a net transmitter of information over the entire sample period, strengthening its case for being the most influential market among others. Japan and India also display consistent behaviour in terms of spillover activity as they remain the net recipients of

Figure 4: Net Directional Spillovers



Notes: (a) The figure presents time evolution of net directional spillovers of all sample markets computed from block aggregation technique under Diebold-Yilmaz framework.
(b) Dynamic spillover index is obtained using fixed rolling window of 200 weeks.

spillovers over the entire period, with spillovers received intensifying during the times of crisis. This indicates that the two countries are exposed to shocks from international markets, especially during the financial turmoil. Hence, investors should invest in alternative asset classes during turbulent times that are relatively safe. Net spillover dynamics of the rest of the stock markets experience different phases as they transition between being the net transmitter and net recipient of spillover, indicating asymmetry in the spillovers. Behaviour of majority of the markets alters especially during the turmoil periods indicating radical change in investor's sentiments. Australia and Luxembourg become source of spillover during the episode of crisis, while South Korea transitions from being a net transmitter to a net receiver of shocks during crisis.

4.4. Implications for Global Financial Centres Ranking

We now turn to undertake comparative analysis of the ranking of the global financial markets obtained from our empirical results of financial linkages among equity markets and the GFCI ranking of the competitiveness of the financial centres. Panel A of Table 4 presents ranking based on the results of average conditional correlations of each sample market with other markets (on an average) computed from ADCC model, gross and net spillovers of each market obtained from block aggregation technique under Diebold-Yilmaz framework and ranking of the financial markets based on GFCI 21¹⁰. Our empirical investigation of the linkages among the global financial markets demonstrates that UK is foremost in the information transmission process with the highest net spillovers. This is fairly consistent with the GFCI ranking of competitiveness of the financial centres that rank London (UK) on the top position as a leading financial centre across the globe. However, other markets that rank among top 5 according to GFCI, viz. US (New York), Singapore, Hong Kong and Japan (Tokyo) are relatively not influential in information transmission compared to equity markets of South Korea and Australia that rank low as per GFCI. Panel B of Table 4 shows the pairwise correlation (Spearman's rho) between the ranking of financial centres based on empirical results and GFCI. Correlation among the ranks obtained under ADCC and Diebold-Yilmaz framework is reasonably high and significant, while the correlation of these with GFCI ranking is substantively low and insignificant. This clearly establishes that the parameters of GFCI ranking, viz. business environment, human capital, infrastructure,

¹⁰ The ranking of the markets presented here differ slightly from the ranking in GFCI 21. As discussed in section 2.1, we arrive at the ranking of financial markets by considering the market/region in which financial centre is based and drop the overlapping ones from our analysis.

financial sector development and reputation may not necessarily imply financial interdependencies with other markets. Indeed GFCI ranking is based on several vital parameters, it overlooks financial linkages with other markets as an attribute based on which underlying strength or dominance of a market can be discerned. We do not argue that the ranking criterion should be based exclusively on financial linkages with the markets, but it should be incorporated in creating the index as it constitutes an integral component in determining the competitiveness of a financial system. Hence, the GFCI ranking should be reviewed in light of the interdependencies among the global financial centres.

Table 4: Global Financial Markets Ranking

Panel A	ADCC	Gross Spillovers	Net Spillovers	GFCI
AUS	3	4	2	6
CAN	7	6	5	7
CHI	10	11	9	9
GER	9	10	11	11
HKG	5	7	8	4
IND	13	12	12	13
JAP	12	13	13	5
LUX	11	8	3	10
SGP	6	3	6	3
SKOR	1	2	4	12
SWIZ	8	9	10	8
UK	2	1	1	1
US	4	5	7	2
Panel B	Spearman's Rank Correlations			
	ADCC	Gross Spillover	Net Spillover	GFCI
ADCC	1.00			
Gross Spillover	0.91**	1.00		
Net Spillover	0.70**	0.85**	1.00	
GFCI	0.45	0.46	0.31	1.00

Notes: (a) Panel A of the table presents the ranking of global financial markets based on the results of average conditional correlations of each sample market with other markets (on an average) computed from ADCC model, gross and net spillovers of each market obtained from block aggregation technique under Diebold-Yilmaz framework and ranking of the financial markets based on GFCI 21.

(b) The ranking of financial markets based on GFCI 21 and the ranking presented here differ slightly as we have arrived at this ranking by considering the market/region in which financial centre is based and have dropped the overlapping ones from our analysis.

(c) Panel B of the table shows spearman's rank correlation among the rankings of financial markets based on our empirical models and GFCI 21.

(d) ** indicates significance at 0.01 level.

5. Interlinkages among Moments

In this section, we assess linkages among returns and higher moments i.e. conditional volatility and conditional skewness so as to investigate risk-return relationship in the global financial markets which has important implications for optimal portfolio selection and risk management. Similar technique as adopted in Section 4.2.2 is employed to examine spillovers across moments by aggregating markets.

Table 5: Spillovers among Moments

	r	v	s	Contri from
r	95.04	3.73	1.23	4.96
v	55.76	41.69	2.55	58.31
s	3.50	7.36	89.14	10.86
Contri to	59.26	11.09	3.77	24.71

Notes: (a) r, v and s denotes return, conditional volatility and conditional skewness

(b) The 3×3 matrix in the table presents spillovers among return and higher order moments obtained by applying block aggregation technique of Greenwood-Nimmo et al. (2015) under Diebold-Yilmaz framework.

(c) The diagonal elements of the matrix represent within moment spillovers and off-diagonal elements represent cross-moment spillovers.

(d) The last column of the table, ‘contri from’, represents spillovers *from* other moments to moment *i*.

(e) The last row of the table, ‘contri to’, represents spillovers *to* other moments from moment *i*.

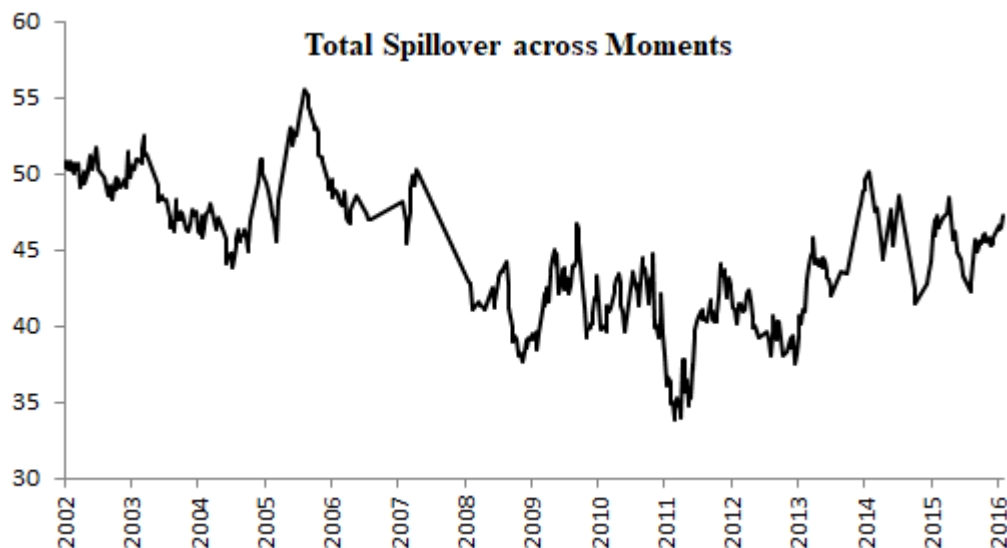
(f) The bottom right corner of the table is the total spillover index.

(g) All values are measured in percentage units.

3 × 3 connectedness matrix presenting the connectedness among group of returns, volatility and skewness, aggregated over all 13 markets is reported in Table 5. The prime diagonal represents within moment spillovers and off-diagonal elements represent cross-moment spillovers. The last column of the matrix labelled ‘contribution from others’ sums the directional spillovers to moment *i* from other two moments and the last row labelled ‘contribution to others’ represents the directional spillover from moment *i* to other two moments in the model. The total spillover index (shown as the bottom right of the matrix) is around 25 percent indicating that information transmission across moments is quite limited. Results demonstrate return spillovers of high magnitude onto risk (primarily volatility), while almost negligible risk spillover onto returns indicating that risk does not impact returns rather is strongly impacted by returns. This is contrary to the premise that returns are explained by risk, which is the cornerstone of the asset pricing theories including Capital Asset Pricing Theory (CAPM) of Sharpe (1964) and Lintner (1965) and thus questions the validity of these theories.

The time evolution of total spillover index shows downward trend in the total spillovers across risk and return during the crisis (refer figure 5). This explains the tendency of risk-return relationship, expressed in a linear form, to break down during the times of high uncertainty period due to large jumps in the asset prices.

Figure 5: Total spillover across moments

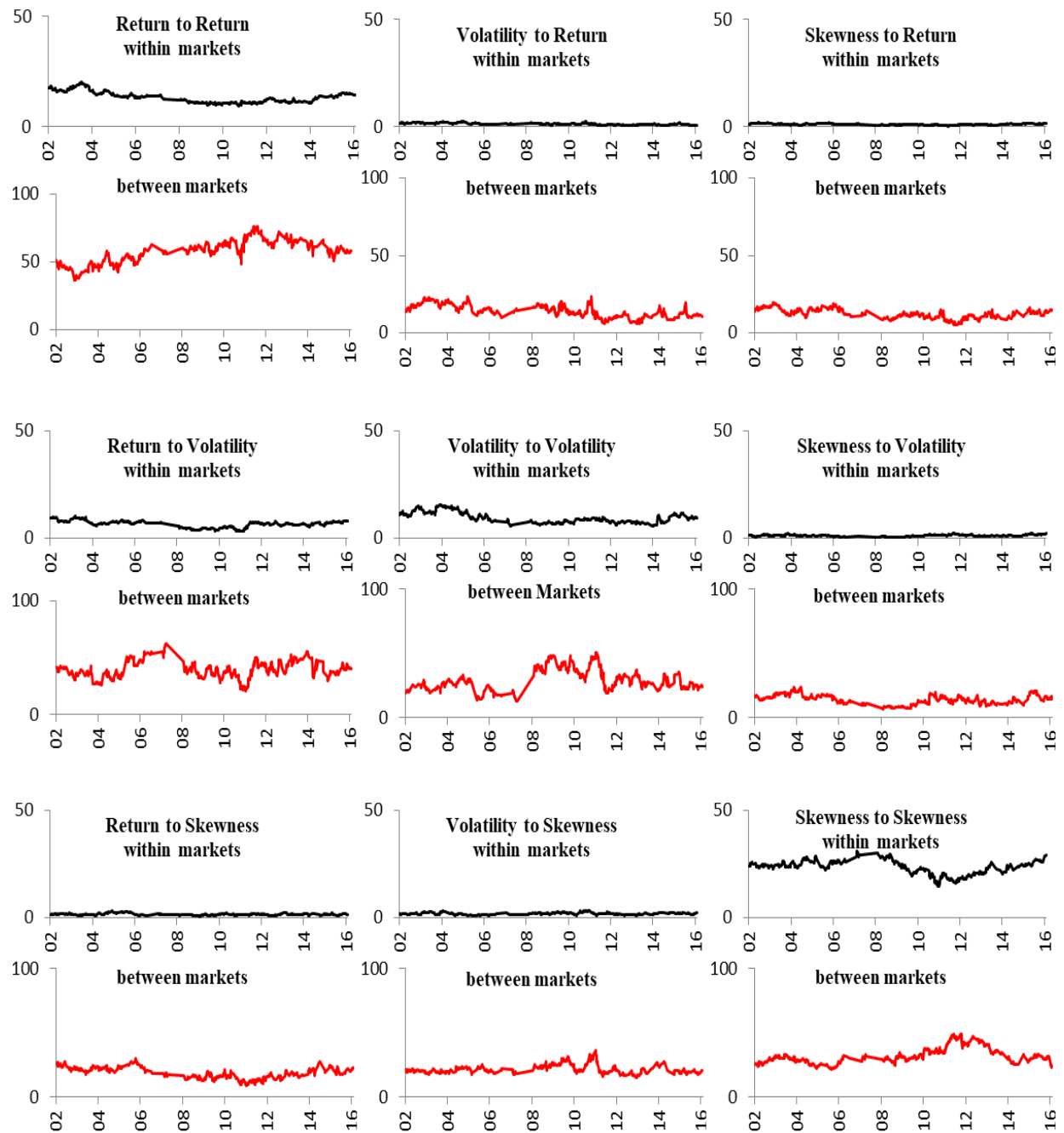


Notes: (a) The figure illustrates evolution of total spillover among return and its moments within the global financial markets, computed from block aggregation technique under Diebold-Yilmaz framework.
 (b) Dynamic spillover index is obtained using fixed rolling window of 200 weeks.

Pairwise directional spillovers across moments are decomposed into within-market effect and cross-market effect to examine the relative importance of idiosyncratic and systematic effects in the moments. Figure 6 presents the evolution of directional spillovers across moments disaggregated into within and cross market effect. Panel A, B and C of figure 6 display the spillovers onto return, volatility and skewness, respectively with upper plot in each panel representing the within-markets effect and lower plot representing between-markets effect. First considering the spillovers to returns, we find that the level of within market effect is substantially lower as compared to the cross-market effect (refer Panel A of figure 6). This indicates that systematic effect driven by common factors is more prevalent in spillovers onto returns as compared to idiosyncratic effect. Surge in the cross-market effect among returns in the wake of Global Financial Crisis and Eurozone Debt Crisis is evident from the results signifying return spillovers across markets intensify during turbulent period. On the other hand, risk spillovers captured by volatility and skewness spillover onto returns drop during

the crisis period, pointing to the weakening of risk-return relationship. Spillovers to volatility exhibit similar time variations as return spillovers onto volatility across markets plunge during the turmoil period (refer Panel B of figure 6). A clear demarcation in spillovers to volatility within and cross market is visible as idiosyncratic effect remains almost negligible. Further, own-variable cross-market effect in volatility soars during the crisis period highlighting that systematic risk intensifies in the times of stress disrupting all markets simultaneously. Now considering return and volatility spillovers onto skewness, we find that within market effect is almost absent (refer Panel C of figure 6). Weakening of risk-return relationship is also evident from the slight decrease in the cross-market effect from return spillovers onto skewness. In addition, the level of own-variable idiosyncratic effect in skewness is distinctly higher as compared to that in return and volatility and also stronger as the cross-market effect in skewness. However, own-market effect among skewness decreases as the cross-market effect increases during crisis period, indicating the risk of coordinated crash of equity prices in the times of stress.

Figure 6: Directional Spillovers across Moments



Notes: (a) The figure illustrates time evolution of directional spillovers among moments computed from block aggregation technique under Diebold-Yilmaz framework.

(b) Panel A (top row), Panel B (middle row) and Panel C (bottom row) display spillovers onto returns, conditional volatility and conditional skewness, respectively.

(c) The upper plot in each panel depicts within-market effect and the lower plot cross-market effect.

(d) Dynamic spillover index is obtained using fixed rolling window of 200 weeks.

6. Summary and Conclusion

In this paper, we investigate dynamic interdependencies among major global financial markets (Australia, Canada, China, Germany, Hong Kong, India, Japan, Luxembourg, Singapore, South Korea, Switzerland, UK and US) over the period January 1999 to April 2017 by examining their conditional correlations and risk and return spillovers. We first employ ADCC model to investigate the asymmetric dynamics of equity market correlations, which is a stylized fact about financial market interdependencies. ADCC results reveal that South Korea and UK has highest conditional correlation with other global financial markets, while India and Japan have the lowest. Block aggregation technique under Diebold-Yilmaz framework proposed by Greenwood-Nimmo et al. (2015) is then employed to examine information transmission across the financial markets that channel through return as well as risk captured by conditional volatility and conditional skewness. Results demonstrate that equity markets of UK and South Korea are the two largest gross transmitter and receiver of information spillovers, while India and Japan are least influential in transmitting spillovers as well as least responsive to the information originating in the major global financial markets thereby reinforcing ADCC model results. The time evolution of total spillover across the global financial markets indicates marked increase in spillover effects since the outset of Global Financial Crisis in 2008 which remain pronounced till the end of Eurozone Debt Crisis in 2012, revealing the contagious nature of these events. Spillovers across markets fall thereafter with the unwinding of the stress period and beginning of global economic recovery. Further, low within market spillovers indicate that financial markets linkages have limited exposure to idiosyncratic effects as they are largely driven by common global economic factors. The systematic effects in these markets intensify during the turmoil period. Based on the empirical results, one can infer the underlying dominance or strength of a financial market. UK seems to emerge as the dominant market in the global financial system as it is the lead net transmitter of information to other markets along with displaying the highest gross spillover activity. Australia and South Korea are other major sources of spillover indicating that they also exert substantial influence on other markets, while India and Japan are least dominant among all the sample markets.

We undertake comparative analysis of financial markets ranking based on our empirical results and GFCI 21. Weak and insignificant correlation is found between the GFCI ranking and ranking based on information transmission, implying that financial interlinkages among the markets may not necessarily reflect in the parameters on which GFCI ranking is based.

Indeed GFCI ranking is derived from several vital parameters, it overlooks information linkages with other markets as an attribute based on which underlying strength or dominance of a market can be discerned. Hence, the existing system of GFCI ranking should be reviewed in light of the interdependencies among the global financial centres as it constitutes a critical component for determining the competitiveness of a financial system.

We also ascertain interlinkages among return, conditional volatility and conditional skewness by aggregating the sample markets through block aggregation routine under Diebold-Yilmaz framework. This provides us useful insights regarding risk-return relationship in the global financial markets. Results reveal return spillovers of high magnitude onto risk (primarily volatility), while almost negligible risk spillover onto returns indicating that return has greater role to play in volatility prediction than vice-a-versa. This challenges the validity of asset pricing models as they hinge on the premise that returns are explained by risk and may not accurately uncover the relationship between return and risk. In this regards, autoregressive or univariate models are superior in modelling return generating process compared to traditional asset pricing models, including CAPM. On the other hand, augmented GARCH specifications will be more relevant for volatility prediction and standard GARCH model for skewness forecasting.

Our results provide evidence of drop in the total spillovers across risk and return in the wake of crisis, pointing to the tendency of risk-return relationship to break down during the times of high uncertainty period due to large jumps in the asset prices. Further, we find that systematic factors are more prevalent in own-variable spillovers among returns and volatility as compared to idiosyncratic factors as is indicated by high level of cross-market effect relative to within-market effect. However, own-variable spillovers among skewness display distinct behaviour as idiosyncratic effect is relatively strong implying that large jumps or crashes in the equity prices are propelled by market-specific factors. The own-variable systematic effect in return, volatility and skewness amplifies during crisis which underscores the role of common global factors in propagating risk and return across the global financial markets in the stress times.

Our empirical findings have important implications for international investors who seek to manage their portfolio and exposure to risk through diversification. Equity markets of India and Japan particularly provide considerable diversification opportunities to international investors. While these markets confer substantial gains during tranquil period, their exposure

to systematic effects intensifies during the crisis period suggesting that investors should invest in alternative asset classes during turbulent times that are relatively safe. The study also has relevance for policymakers as it suggests that coordinated response is required at times of uncertainty through policy synchronization so as to insulate the economy from external headwinds and ensure financial stability.

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